DATA SCIENCE BOOTCAMP #15 | CAPSTONE FINAL PRESENTATION | APRIL 2026

Supply Chain Logistics Analysis

Diagnosing lead-time bottlenecks in a multi-echelon logistics network

Presented by Hazaimi Hamdani and Intan Nurhazwani

4,899
shipment records analysed

97%
of shipments via PORT04

1.32 days
Random Forest RMSE (R^2 0.29)

Dataset: Kaggle, Supply Chain Logistics Problem | Dashboard: supply-chain-logistics-analysis.netlify.app

Problem Statement & Objectives

Reactive, siloed supply chains turn small demand swings into costly inventory and delivery problems

Bullwhip effect

Silos between supply chain stages turn demand fluctuations into overstocking at warehouses.

Excess capital costs

Inefficient routing and unpredictable lead times tie up capital in dead stock and overflow storage.

Delivery delays

Unpredictable port arrivals and transit times force excess buffer inventory across the network.

Research objectives

1. Identify bottlenecks in the logistics network using statistical analysis
2. Build a machine-learning model to predict Total Lead Time
3. Deliver actionable, evidence-backed business recommendations

Dataset & Methodology

Kaggle Supply Chain Logistics Problem: the OrderList table, from a 7-table relational schema

4,899
records analysed after cleaning

9,215
raw OrderList records

14
fields in OrderList

7
tables in source schema

1

Data cleaning

Removed duplicates, median-imputation safeguard, trimmed Weight outliers (1st to 99th percentile).

2

EDA

Distributions, correlation heatmap, port, plant and carrier profiles.

3

Hypothesis testing

One-way ANOVA, Welch t-test, Pearson correlation, Tukey HSD.

4

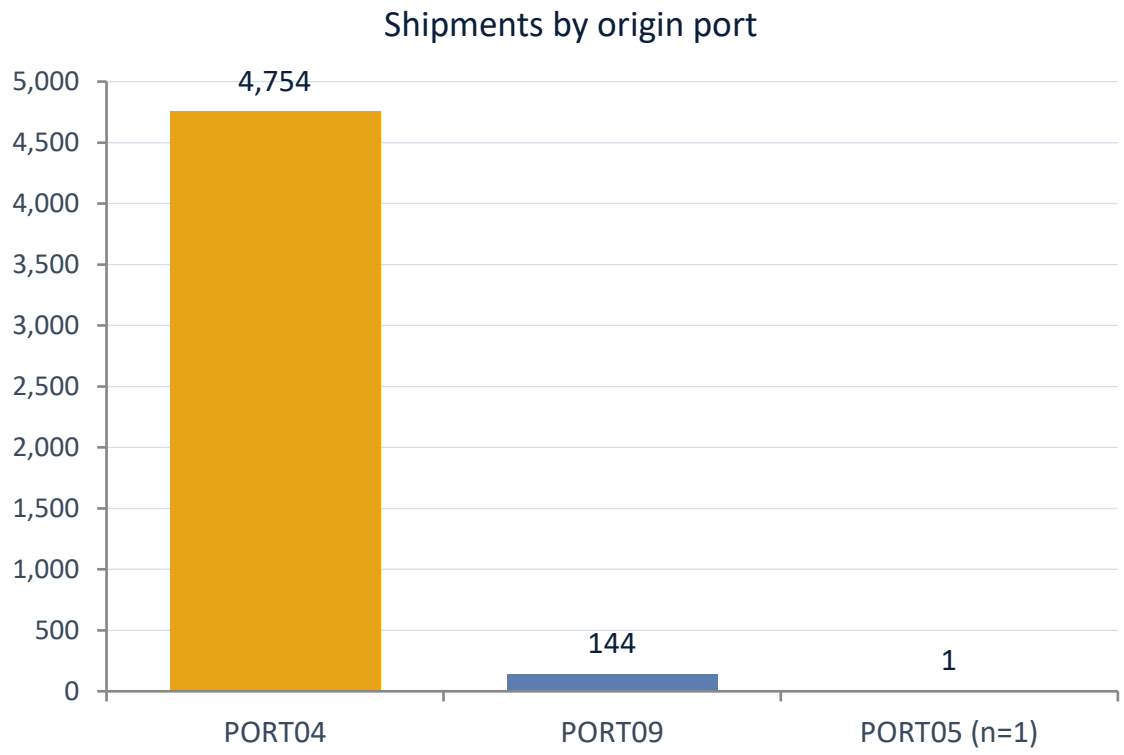
ML modelling

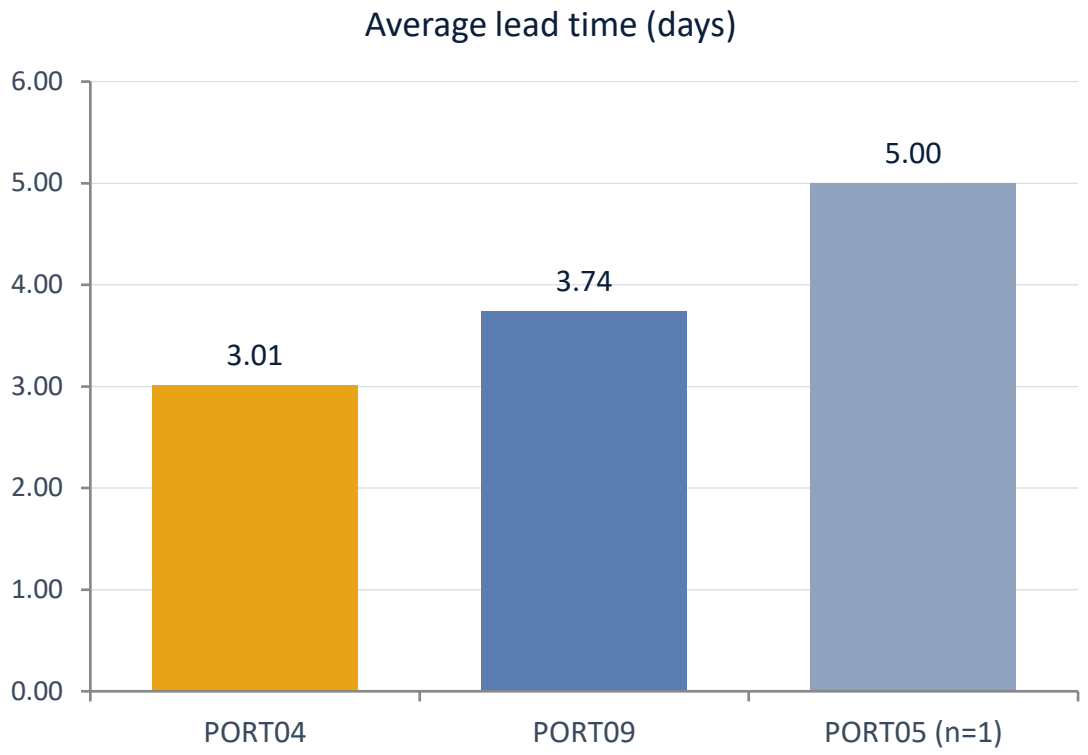
Linear Regression, Random Forest, OLS and XGBoost on an 80/20 split.

Sampling note: 5,000 records were retained per the project size guideline. The source Order Date field holds a single date, so the analysis is cross-sectional.

PORT04 Concentration Risk

97% of shipments depend on one origin port, creating a single point of failure



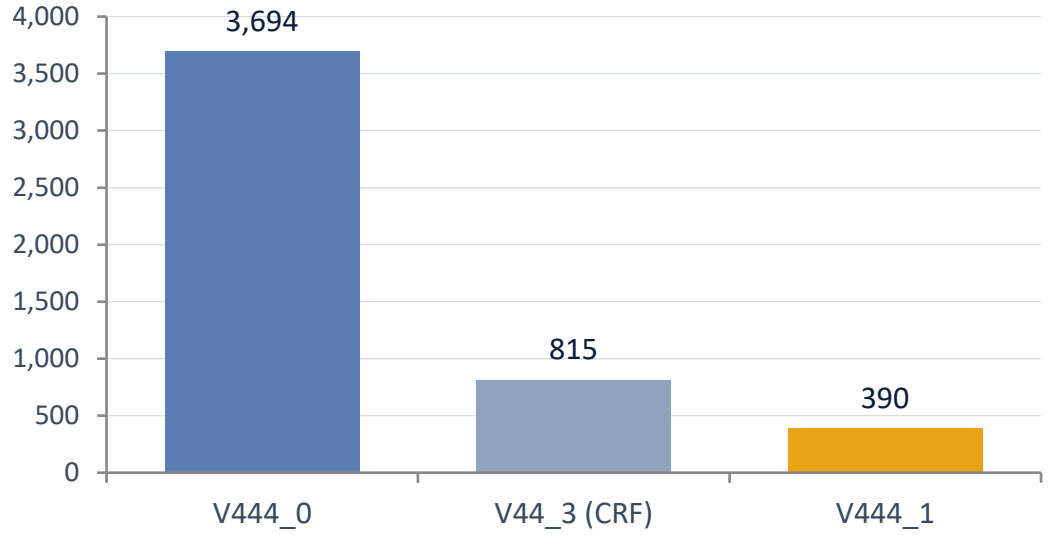


Port significantly affects lead time (ANOVA F = 16.08, p = 1.09e-07). PORT05 has a single record and is too thin to interpret.

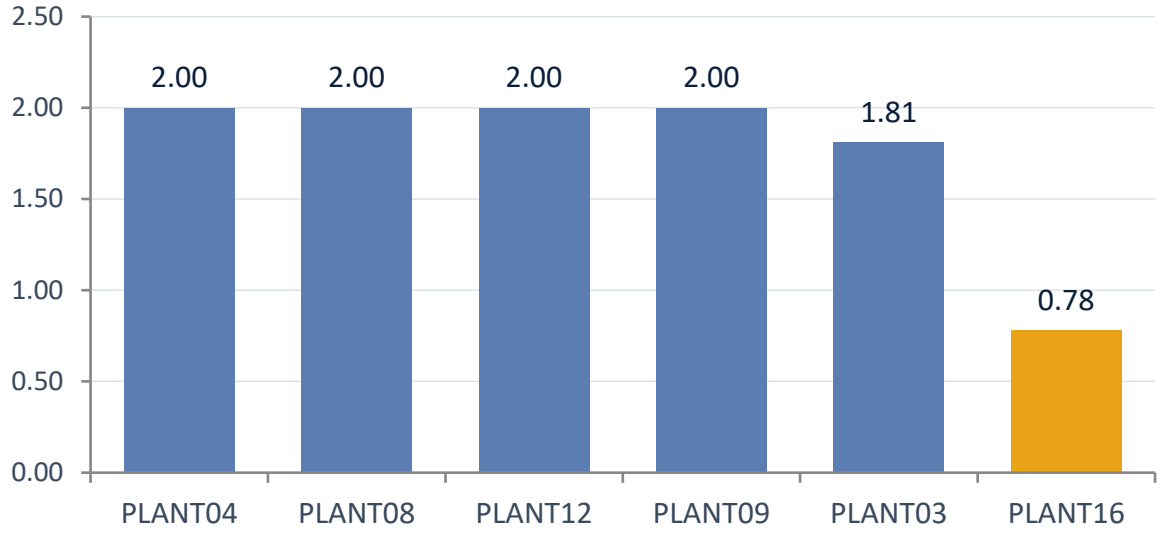
Carrier and Plant Drivers

Carrier choice and plant both significantly affect transit processing time (TPT)

Shipments by carrier



Average TPT by plant (days)



ANOVA F = 754.0 ($p < 0.001$). OLS links V444_1 to about 0.44 fewer lead-time days than V444_0, but it carries only 8% of volume.

PLANT16 is significantly faster than PLANT03, 09 and 12 (Tukey $p < 0.001$). PLANT09 delay is above the network ($t = 8.04$, $p = 3.0e-06$).

Statistical Evidence

Six tests support the bottleneck findings (significance level 0.05)

Test	Statistic	p-value	Reading
Carrier vs TPT (one-way ANOVA)	F = 754.00	3.46e-286	Carrier affects TPT
Port vs lead time (one-way ANOVA)	F = 16.08	1.09e-07	Port affects lead time
PLANT09 vs rest (Welch t-test, one-sided)	t = 8.04	3.03e-06	PLANT09 delay is higher
Tukey HSD, TPT by plant	PLANT16 vs 03, 09, 12	< 0.001	PLANT16 is faster
Pearson: TPT vs lead time	r = 0.32	< 0.001	Moderate positive link
Pearson: Weight vs lead time	r = 0.067	< 0.001	Significant but weak

PLANT09 delay is the derived metric Total Lead Time minus TPT. Its high correlation with lead time (r = 0.89) is built in by definition and is not treated as a finding.

Predictive Model: Lead Time

Random Forest predicts lead time within about 1.1 days on average, but explains only 29% of variance

1.32 days

RMSE

1.08 days

MAE

0.29

R^2 (variance explained)

Model	RMSE (days)	R^2
Random Forest (champion)	**1.32**	**0.29**
Linear Regression	1.43	0.17

XGBoost and OLS were also evaluated. Their figures will be re-run into one reproducible leaderboard before publication.

Top Random Forest drivers

1. Unit quantity
2. Weight
3. Service level (DTD)
4. Carrier V444_1

The modest R^2 reflects factors outside the data, such as real-time congestion, weather and seasonality.

Business Recommendations

Three evidence-backed actions, each with its limits stated

1

Reduce single-port dependence

Action: Pilot routing selected non-urgent shipments via PORT09 and track lead time.

Evidence: 97% of volume uses PORT04. PORT09 averages 0.7 days longer, so the aim is resilience, not speed.

Resilience pilot

2

Test carrier consolidation

Action: Pilot more volume with V444_1, then add TPT and on-time SLA clauses. Exclude CRF carrier V44_3.

Evidence: Lowest TPT in the ANOVA and about 0.44 fewer lead-time days in OLS, but only 8% of volume.

Pilot first

3

Review plant workload and buffers

Action: Rebalance workload toward faster plants and size safety stock on lead time.

Evidence: TPT is 2.0 days at four plants vs 0.78 at PLANT16 (Tukey $p < 0.001$).

Validate with cost data

Limitations & Next Steps

Limitations

- Single order date: no time-series or seasonality
- Small groups (PORT05 n = 1, some plants)
- R^2 of 0.29: external factors not captured
- Cost impact not quantified

Next steps

- Use freight-rate, capacity and cost tables to size savings
- Add dated data for time-based validation
- Re-run all models in one cross-validated pipeline
- Build a carrier cost-and-performance scorecard

Thank you

Hazaimi Hamdani | Intan Nurhazwani | Data Science Bootcamp #15